25900315 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-BCA (HONOURS) Regular EXAMINATION October 2025

Third SEMESTER

Core Course(CC) - MG3CCRBCA201 - DATABASE MANAGEMENT SYSTEMS

(2024 ADMISSION ONWARDS)

Duration: 2 Hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Very Short Answer Questions

Answer all questions

Each question carries **2** marks

1. Can a primary key be NULL or duplicate? Justify. [An] / [CO1]
2. Define TRC. [U] / [CO2]
3. State the reflexivity property of functional dependencies. [K] / [CO3]
4. Write the use of Cassandra. [A] / [CO4]
5. What is the role of a lock manager in concurrency control? [U] / [CO3]

[2x5 = 10]

Part B

Short Answer Questions

Answer any **5** questions

Each question carries **6** marks

6. Analyze the purpose of DBMS in managing large amounts of data. Why is it preferred over file-based data storage? [An] / [CO1]
7. Summarize serializability importance in transaction management. [U] / [CO3]

8.	Write SQL queries to demonstrate the use of logical operators (AND, OR, NOT) in filtering records.	[A] / [CO2]
9.	Compare column family and graph data models of NoSQL databases.	[An] / [CO4]
10.	Discuss about dirty read, nonrepeatable read and phantom problem in a transaction.	[U] / [CO3]
11.	Explain the concept of FULL OUTER JOIN in SQL with proper syntax, example tables, and expected output. Also, discuss how it differs from LEFT and RIGHT OUTER JOINS.	[U] / [CO2]
12.	Analyze the importance of primary keys in the relational model with relevant examples.	[An] / [CO2]

[6x5 = 30]

Part C

Essay Questions

Answer any **2** questions

Each question carries **15** marks

13.	Explain the notations of ER diagram.Create an Entity-Relationship (ER) diagram for a university database that includes entities like 'student', 'course', and 'instructor'. Define suitable relationships and attributes for each entity.	[A] / [CO2]
14.	Define a data model. Explain the different types of data models in DBMS with diagrams and examples.	[K] / [CO1]
15.	With suitable steps and examples, explain how to compute the closure of a set of functional dependencies.	[A] / [CO3]

[15x2 = 30]